

Reactome Analysis Result

04 February, 2025

Introduction

This report summarizes the pathway analysis result created by the **Reactome Analysis Service**. For more information, visit <https://www.reactome.org>.

In this reports, pathways found differentially expressed at an adjusted p-value ≤ 0.05 are considered to be significantly regulated.

Analysis Overview

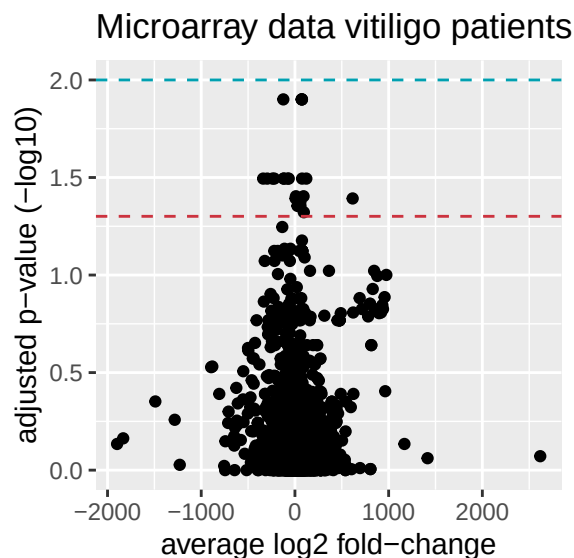
- Number of datasets analysed: 1
- Reactome version: 91
- Disease pathways were included

Dataset summaries

Microarray data vitiligo patients lesional and non-lesional skin:

- 2650 pathways
- 15681 fold changes for genes / proteins

Pathway analysis



These volcano plots summarize the pathway results for every dataset. Every point is one pathway. The x-axis represents the average fold-change of all genes / proteins within that pathway. The y-axis represents the p-value where “higher” values are more significant ($-\log_{10}$ transformation). The red line represents $p = 0.05$, and the blue line $p = 0.01$.

Note

Since this result only contains a single datasets, some analyses are not included in this report.

Top-regulated pathways

Amplification of signal from the kinetochores (R-HSA-141424)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.013
Av. FC	74.438
N Genes	76

Amplification of signal from unattached kinetochores via a MAD2 inhibitory signal (R-HSA-141444)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.013
Av. FC	74.438
N Genes	76

Mitotic Spindle Checkpoint (R-HSA-69618)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.013
Av. FC	71.373
N Genes	91

Protein lipoylation (R-HSA-9857492)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.032
Av. FC	121.303
N Genes	9

Mitotic Telophase/Cytokinesis (R-HSA-68884)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.032
Av. FC	76.174
N Genes	9

Cohesin Loading onto Chromatin (R-HSA-2470946)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.039
Av. FC	92.124
N Genes	7

Resolution of Sister Chromatid Cohesion (R-HSA-2500257)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.039
Av. FC	10.642
N Genes	104

Formation of ATP by chemiosmotic coupling (R-HSA-163210)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.041
Av. FC	615.991
N Genes	17

Mitotic Prometaphase (R-HSA-68877)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.041
Av. FC	6.883
N Genes	167

Mitochondrial translation (R-HSA-5368287)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.041
Av. FC	49.598
N Genes	93

Mitochondrial translation termination (R-HSA-5419276)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.042
Av. FC	57.728
N Genes	87

Mitochondrial translation initiation (R-HSA-5368286)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Up
FDR	0.044
Av. FC	58.61
N Genes	87

Melanin biosynthesis (R-HSA-5662702)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Down
FDR	0.013
Av. FC	-124.068
N Genes	4

Maturation of spike protein (R-HSA-9683686)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Down
FDR	0.032
Av. FC	-220.216
N Genes	4

Chondroitin sulfate/dermatan sulfate metabolism (R-HSA-1793185)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Down
FDR	0.032
Av. FC	-116.411
N Genes	48

Glycosaminoglycan metabolism (R-HSA-1630316)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Down
FDR	0.032
Av. FC	-66.513
N Genes	121

Gluconeogenesis (R-HSA-70263)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Down
FDR	0.032
Av. FC	-295.284
N Genes	24

Regulation of MITF-M-dependent genes involved in pigmentation (R-HSA-9824585)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Down
FDR	0.032
Av. FC	-339.081
N Genes	29

Metabolism of carbohydrates (R-HSA-71387)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Down
FDR	0.032
Av. FC	-76.749
N Genes	270

EPH-ephrin mediated repulsion of cells (R-HSA-3928665)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Down
FDR	0.032
Av. FC	-239.527
N Genes	36

Heparan sulfate/heparin (HS-GAG) metabolism (R-HSA-1638091)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Down
FDR	0.032
Av. FC	-121.462
N Genes	55

Diseases of carbohydrate metabolism (R-HSA-5663084)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Down
FDR	0.044
Av. FC	27.077
N Genes	32

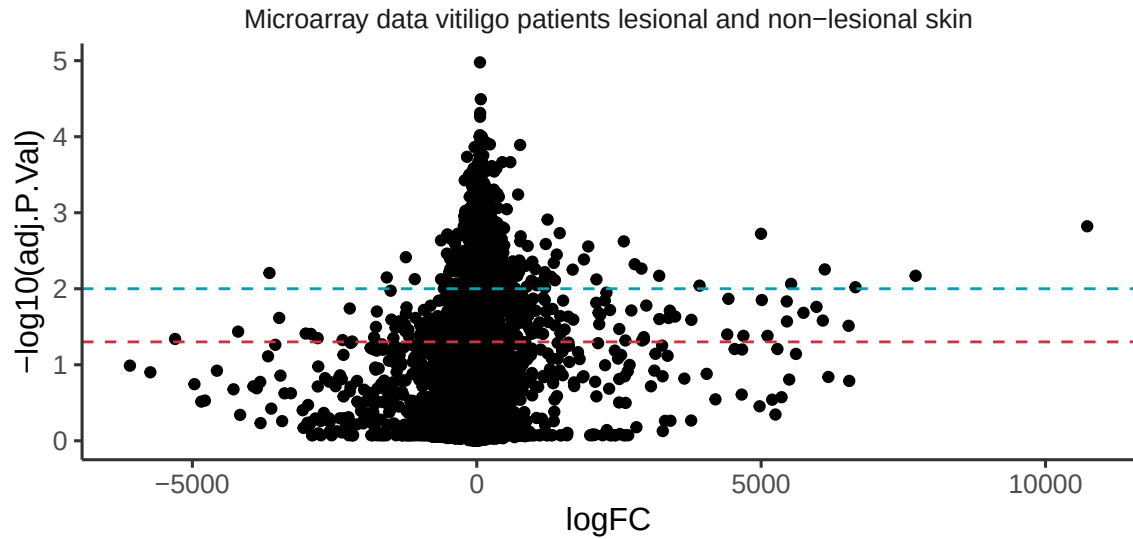
Ephrin signaling (R-HSA-3928664)

	Microarray data vitiligo patients lesional and non-lesional skin
Regulation	Down
FDR	0.048
Av. FC	94.793
N Genes	12

Protein / Gene level results

Classical differential expression analysis performed on the genes / proteins of every dataset.

Note: Depending on the gene set analysis method used, the approach used to assess differential expression at the gene / protein level may vary to the approach used for the pathway level.



Top up-regulated protein/genes

Microarray data vitiligo patients lesional and non-lesional skin

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
ILMN_1694502	56.35051	81.35729	9.171453	0e+00	0.0000105	-1.948859
ILMN_2274180	71.68176	89.92456	8.403108	0e+00	0.0000321	-2.087188
ILMN_1702526	59.88918	105.29512	8.063919	0e+00	0.0000487	-2.155679
ILMN_1805828	58.38739	72.89827	7.900119	0e+00	0.0000547	-2.190534
ILMN_1801467	56.36424	86.00054	7.586685	0e+00	0.0000955	-2.260642
ILMN_1807095	49.53356	86.45940	7.481682	0e+00	0.0000998	-2.285169
ILMN_1739770	89.86509	100.84677	7.435478	0e+00	0.0000998	-2.296132
ILMN_2406468	159.98017	226.17814	7.315608	1e-07	0.0001184	-2.325067
ILMN_1764415	134.90898	121.79806	7.224974	1e-07	0.0001261	-2.347424
ILMN_1788701	41.42791	50.17429	7.176389	1e-07	0.0001261	-2.359582

Top down-regulated protein/genes

Microarray data vitiligo patients lesional and non-lesional skin

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
ILMN_1675937	-36.39725	72.32549	-7.039455	1.0e-07	0.0001371	-2.394506
ILMN_1657550	-170.45415	206.84168	-6.807976	2.0e-07	0.0001840	-2.455805
ILMN_1815859	-55.46638	110.19185	-6.447350	6.0e-07	0.0002607	-2.557170
ILMN_1756542	-49.06243	83.12784	-6.363255	7.0e-07	0.0002607	-2.581866
ILMN_2278729	-132.00959	267.23635	-6.218983	1.1e-06	0.0003150	-2.625187

Identifier	logFC	AveExpr	t	P.Value	adj.P.Val	B
ILMN_1737162	-76.46183	69.58952	-6.189266	1.1e-06	0.0003225	-2.634261
ILMN_2409876	-46.17772	84.38194	-6.184425	1.2e-06	0.0003225	-2.635744
ILMN_1697773	-22.07128	62.48418	-6.143678	1.3e-06	0.0003345	-2.648282
ILMN_1758337	-210.96946	287.11671	-6.072681	1.6e-06	0.0003744	-2.670362
ILMN_1792820	-32.37799	88.87997	-5.999251	1.9e-06	0.0004172	-2.693513